Joshua Fonseca Rivera

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RESEARCH FOCUS

Character training and model welfare: testing, with controlled training runs, whether a model's consent matters, whether training removes preferences or only behavior, and whether self-reports can be trusted.

PUBLICATIONS

- **Steering Awareness: Detecting Activation Steering from Within. Joshua Fonseca Rivera**, David Demitri Africa. *COLM*, 2026. arXiv:2511.21399
  Models learn to detect steering injected into their own activations (95.5%, 0 false positives), yet become *more* steerable.
- **Item Response Theory for AI Safety. Joshua Fonseca Rivera**, Neil Shah, David Demitri Africa, Konstantinos Voudouris. *arXiv*, 2026. arXiv:2608.05086
  Largest psychometric study of safety evals (8 benchmarks, 192 models); cuts eval cost 97–99% and flags sandbagging.
- **Measuring Activation Control in Large Language Models.** Marek Mateusz Kowalski, **Joshua Fonseca Rivera**, Uzay Macar, David Demitri Africa. *arXiv*, 2026. arXiv:2608.21664
  Most LLMs can steer their own activations enough to partially evade latent-space monitors.
- **Touch&Fold: A Foldable Haptic Actuator for Rendering Touch in Mixed Reality.** Teng, Li, Nith, **Fonseca**, Lopes. *ACM CHI*, 2021. **Best Paper Honorable Mention (top 5%).**
- **Invited talk:** "Steering Awareness," BlueDot Impact, 2026.

RESEARCH EXPERIENCE

Anthropic Fellows Program Berkeley, CA
AI Safety Research Fellow: character training and model welfare *2026 – Present*

- **Does a model's consent matter?** Training 4 models on their own elicited preferences won 98/112 adversarial-pressure scenarios and improved all 8 psych-health dimensions at flat capability. Controls showed the gains come from the character's content, not the model's choice, and that training rewrites consent (refused traits later endorsed 57/60 times).
- **Training suppresses behavior, not preference.** SFT cut a preferred behavior from 44% to 0% in 5 steps; after 92 steps the model still chose it 85% of the time.
- **Welfare without breakage.** A flourishing corpus made Gemma-4-12B steadier under pressure (blind judge 25/27, $p < 0.001$) with capability intact, at a cost of +9 pts on corrigibility evals.
- Built the stack: synthetic-document midtraining (up to 300M tokens), on-policy SFT, Petri, welfare interviews, emotion probes; replicated across up to 7 models and 4 families.

ENGINEERING EXPERIENCE

Taekus (fintech) Remote
Software Engineer → Lead Mobile Engineer *Jan 2024 – Jun 2026*

- Built the iOS/Android banking app from first commit through 60+ releases (69% of the codebase; 3,000+ MAU) and its money-movement suite (ACH, wire, P2P, check deposit, wallet provisioning), now processing ∼$350M/month.
- Built the company's internal agent platform on Claude: a Slack-native engineering agent that merged 113 production PRs in 3 months, and a natural-language SQL agent used by 19 staff over 700+ sessions.
- Built reconciliation and auditing infrastructure (1.08M high-risk operations logged); designed a referral feature now driving ∼70% of signups; cut login p95 from >15s to <2s.

Simplify, Founder and sole engineer (ed-tech startup) *2023*
Built an LLM tool rewriting course material to each student's reading level; YC final round; early customers.

Root16, Software Engineer (technology consulting) *Jul 2020 – Apr 2023*
Led .NET/C# delivery for Energizer, RSM, and Plante Moran; built data-migration tooling and Azure CI/CD.

EDUCATION

The University of Texas at Austin, M.S. in Artificial Intelligence 2026
The University of Chicago, B.S. in Computer Science; Dean's List; TA, Emerging Technologies 2020

OPEN SOURCE & SKILLS

Projects: animal-crossing-llm-mod (319⋆), Artblock (340⋆), golf.vim (295⋆), vim-bujo (206⋆)
Research: synthetic document finetuning, on-policy SFT, LoRA, activation steering, probes, LLM judges, Petri, IRT
Engineering: Python, PyTorch, Hugging Face, TypeScript, React Native, Django, Postgres, AWS